Exploration of FELLOW trait dataset

The objective of this document is to:

- visually explore the trait database
- understand trait coverage / data gaps
- check for possible inconsistencies

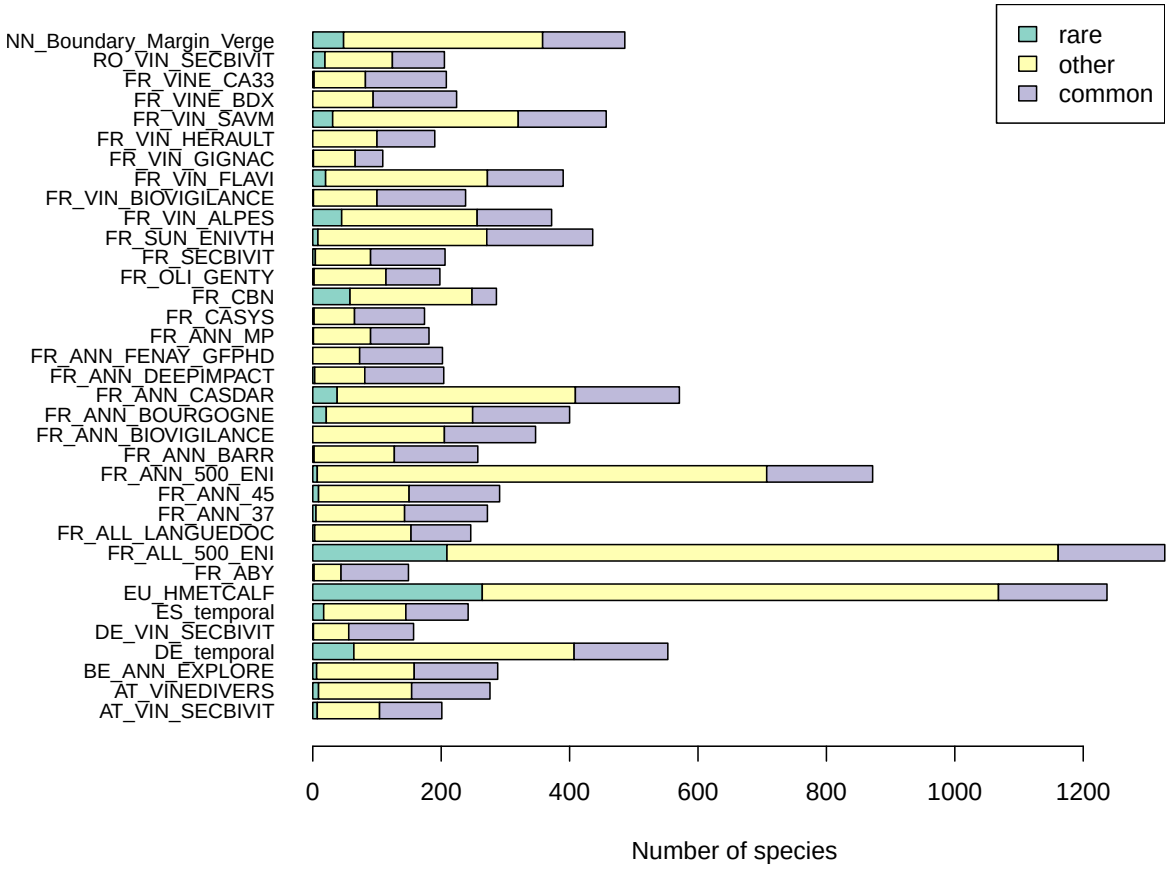
Description of the species list

We compiled the species lists from 35 datasets. After cleaning and harmonization, there were 2416 unique taxa.

```
FAMILY      GENUS    SPECIES SUBSPECIES      TRIBE    VARIETY
    18        328       1875        170          1         24
```

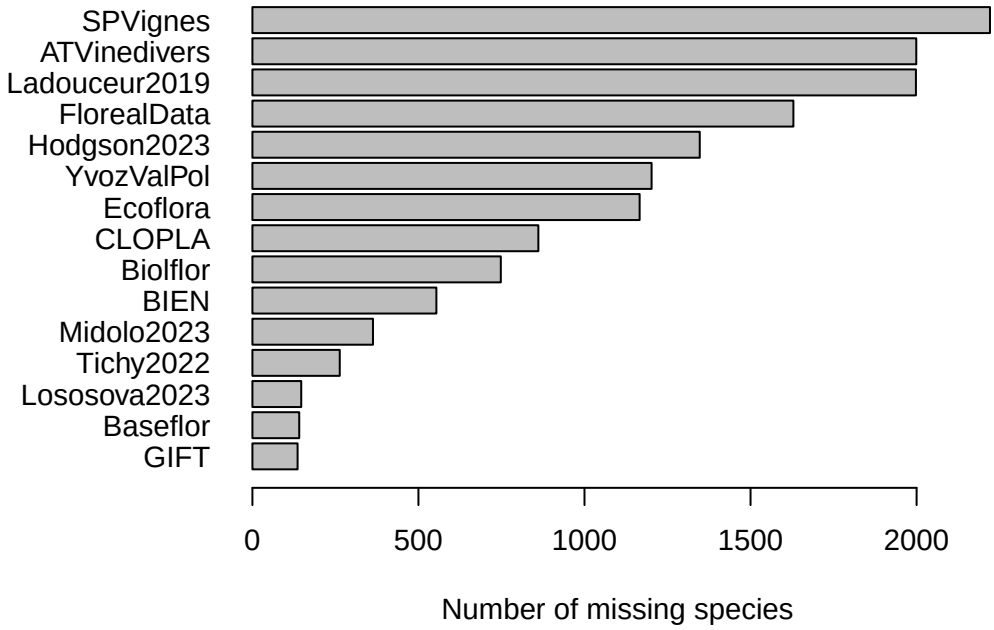
Let's define:
rare: a taxa that occurred only in a singe database
common: a taxa that is listed in 50% of the databases (18 out of 35)

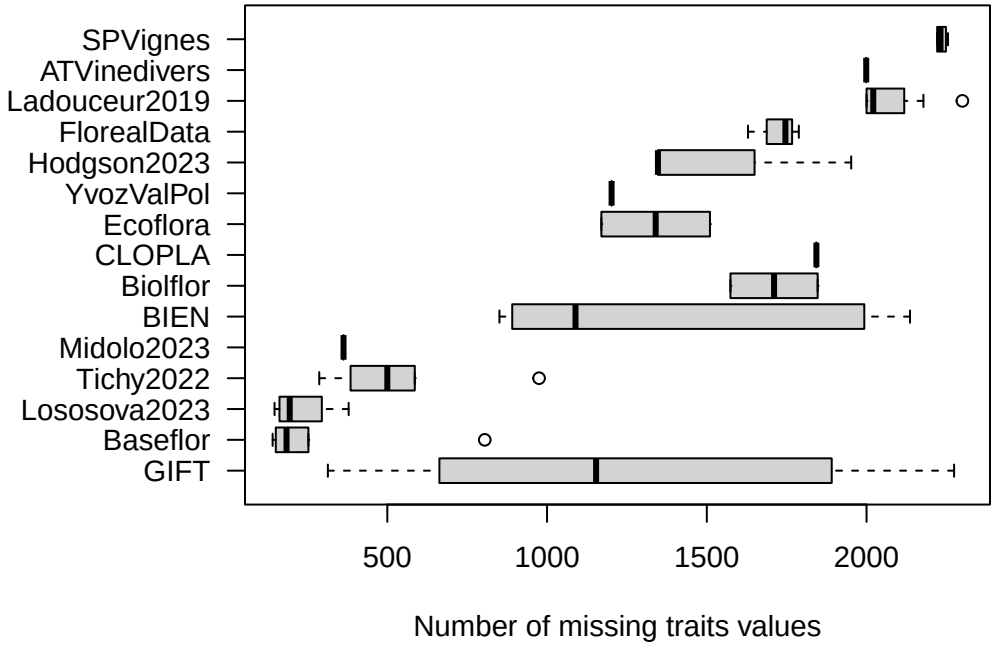
```
sp_class
  rare  other common
   909   1336    170
```

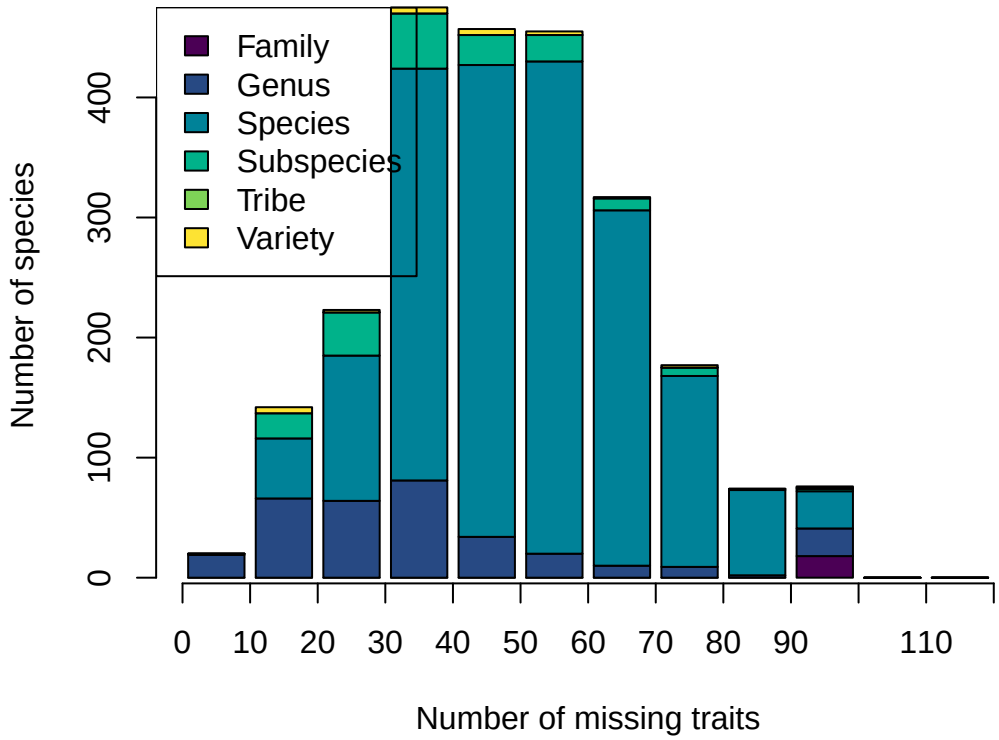


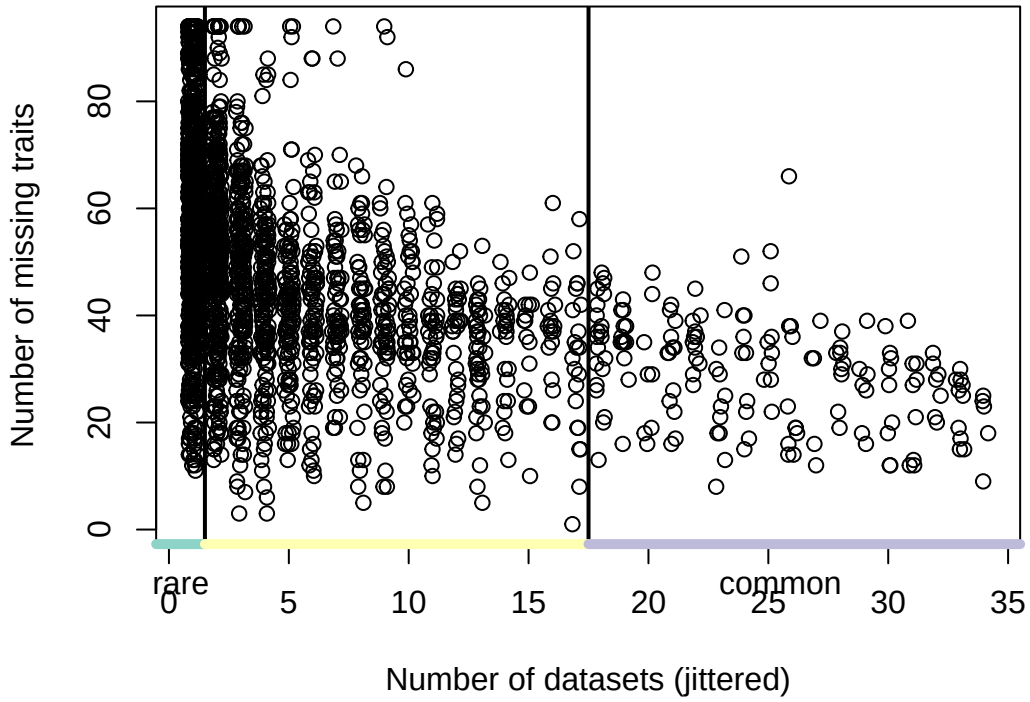
Description of trait databases

So far, we compiled 94 traits for 2416 taxa gathered from 15 trait databases. But there are many missing values.






Family
Genus
Species
Subspecies
Tribe
Variety
Number of species
0
100
200
300
400
0
10
20
30
40
50
60
70
80
90
110
Number of missing traits


Number of missing traits
0
20
40
60
80
rare
common
0
5
10
15
20
25
30
35
Number of datasets (jittered)

Taxa with no or limited trait information (N=53).

```
 [1] "Abies"
 [2] "Acacia"
 [3] "Acalypha ostryifolia"
 [4] "Agropyron"
 [5] "Amaranthaceae"
 [6] "Anagyris"
 [7] "Apiaceae"
 [8] "Arthonia arthonioides"
 [9] "Asparagaceae"
[10] "Aster"
[11] "Boraginaceae"
[12] "Brachythecium rutabulum"
[13] "Brassicaceae"
[14] "Bromeliaceae"
[15] "Bryum dichotomum"
[16] "Cardueae"
[17] "Caryophyllaceae"
[18] "Chaenomeles x superba"
[19] "Chrysanthemum"
[20] "Circaea"
[21] "Cochlearia"
[22] "Cupressus"
[23] "Elymus campestris x repens"
[24] "Ephippiger perforatus"
[25] "Geraniaceae"
[26] "Imbribryum subapiculatum"
[27] "Lamiaceae"
[28] "Lavandula"
[29] "Liliaceae"
[30] "Lilium"
[31] "Marchantia polymorpha subsp. ruderalis"
[32] "Matthiola"
[33] "Milium"
[34] "Moehringia"
[35] "Muhlenbergia"
[36] "Orchidaceae"
[37] "Orchis"
[38] "Paronychia"
[39] "Pelargonium"
[40] "Piptatherum"
```

```
[41] "Poaceae"
[42] "Primulaceae"
[43] "Psoralea"
[44] "Rhizogemma staphylina"
[45] "Riccia"
[46] "Riccia sorocarpa"
[47] "Riccia warnstorfii"
[48] "Roemeria apula"
[49] "Rosaceae"
[50] "Rubiaceae"
[51] "Thapsia"
[52] "Ulmaceae"
[53] "x Triticosecale"
```

Open question:

How to deal with families taxa?

How to deal with missing trait values?

Summary of trait completness

Trait	N database	N taxa	Completness (%)
Dispersal.mode	4	2331	96
Plant.height	8	2284	95
Chorology	1	2275	94
Pollination	1	2265	94
Dispersal.distance	1	2238	93
Flower.color	5	2237	93
Inflorescence	2	2237	93
Lifecycle	2	2184	90
Seed.mass	6	2135	88
Ellenberg.Salinity	1	2129	88
Flowering	2	2059	85
Disturbance.Frequency	1	2053	85
Disturbance.Frequency.herblayer	1	2053	85
Disturbance.Severity	1	2053	85
Disturbance.Severity.herblayer	1	2053	85
Grazing.Pressure	1	2053	85
Mowing.Frequency	1	2053	85
Soil.Disturbance	1	2053	85
Photosynthetic.pathway	1	2039	84
Ellenberg.Light	1	2031	84
Ellenberg.Reaction	1	1998	83
Ellenberg.Nutrients	1	1834	76
Ellenberg.Moisture	1	1830	76
SLA	4	1715	71
Pollination.syndrome	3	1698	70
Flower.UV.reflectance	3	1460	60
Ellenberg.Temperature	1	1441	60
Leaf.area	2	1430	59
Leaf.dry.mass.content	1	1256	52
Floral.symmetry	1	1214	50
Flower.class	1	1214	50
Flower.type	1	1214	50
Nectar.quantity	1	1214	50
Pollen.quantity	1	1214	50
PV.Bees	1	1214	50
PV.Bumblebees	1	1214	50
PV.butterflies	1	1214	50
PV.Hoverflies	1	1214	50
Lifeform	2	845	35

Trait	N database	N taxa	Completness (%)
Flower.length	2	589	24
Clonal.index	2	581	24
lateral.spread	1	575	24
offspring	1	573	24
offspring.wsmall	1	573	24
Flowering.onset	2	553	23
Vegetative.propagation	1	464	19
Leaf.nitrogen.content	1	423	18
Growth.form	1	417	17
Plant.lifespan	1	417	17
Strategy.Grime	1	417	17
Flowering.duration	1	416	17
Seeding.onset	1	416	17
Seeding.season	1	416	17
Clonal.presence	1	384	16
Leaf.carbon.to.nitrogen.content	1	280	12
Seed.dormancy	1	238	10
Strategy	1	174	7
Flower.width	2	165	7
Radial.growth	1	116	5

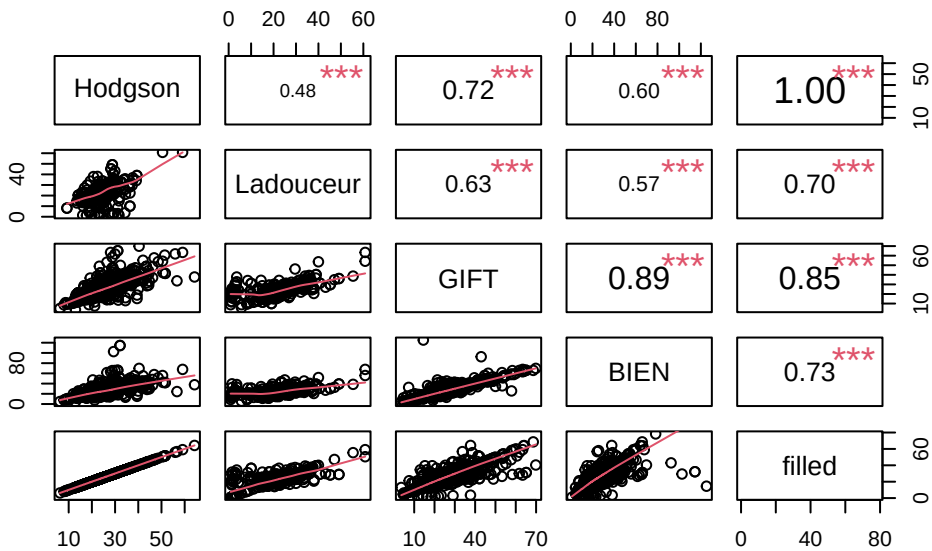
Comparison and merge

SLA

There are four sources of information for Specific leaf area (SLA) : Hodgson et al. 2023 (in mm2/mg), Ladouceur et al. 2019 (in cm2/g), GIFT (in cm2/g), and BIEN (in m2/kg = mm2/mg).

To merge and fill the SLA values (in m2/kg), we used by order of preference:

- Hodgson et al. 2023
- Ladouceur et al. 2019
- GIFT
- BIEN



Number of NAs:

```
Hodgson Ladouceur      GIFT      BIEN    filled
   1347      2060      1019      1018       701
```

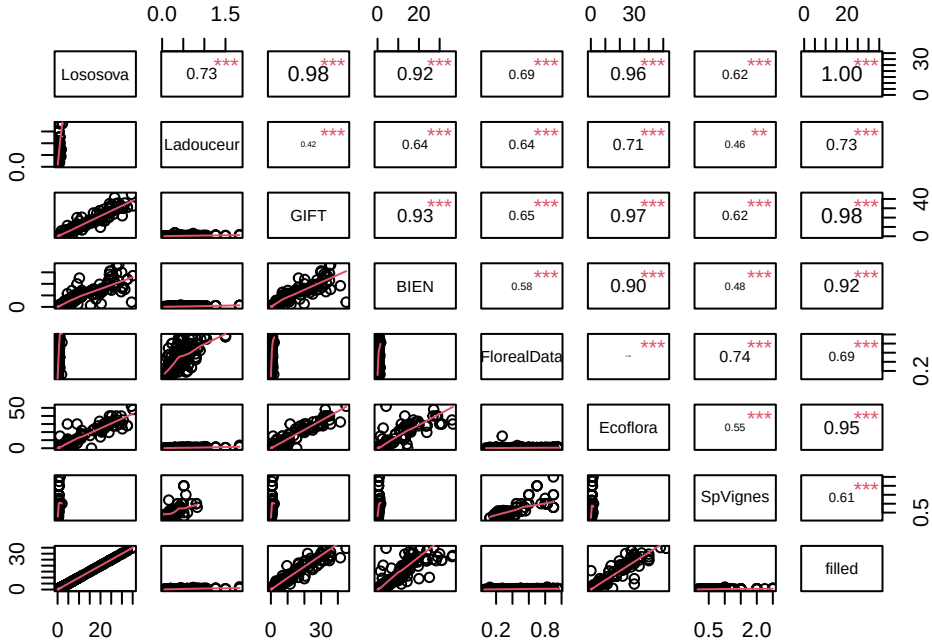
Values are highly correlated, but values from *Ladouceur et al. 2023* are only weakly correlated to other data sources (r<0.65) maybe due to different measuring protocol (?).

Plant height

There are seven sources of information for plant height. SPVignes, FlorealData, and Ladouceur_2019 are limited to small plants (<1m) (no trees) *and FlorealData>100cm must be clean.*

To merge and fill the plant height values (in m), we used by order of preference:

- Lososova et al. 2023
- Ladouceur et al. 2019
- GIFT
- BIEN
- FlorealData
- Ecoflora
- SpVignes



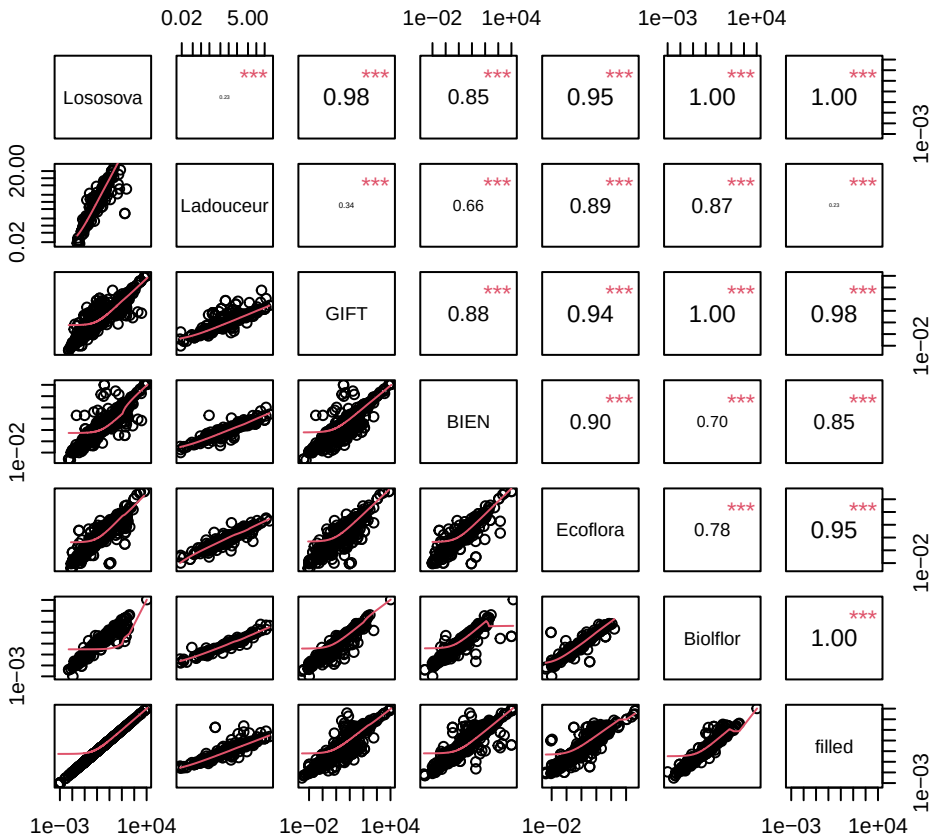
Number of NAs:

```
Lososova   Ladouceur       GIFT       BIEN FlorealData    Ecoflora
     211        2021        663        851        1885        1170
SpVignes      filled
    2255         160
```

Seed mass

There are six sources of information for seed mass. To merge and fill the seed mass plant height values (in mg), we used by order of preference:

- Lososova et al. 2023
- Ladouceur et al. 2019
- GIFT
- BIEN
- Ecoflora
- Biolflor



Number of NAs:

```
Lososova Ladouceur      GIFT      BIEN  Ecoflora  Biolflor    filled
     379      2019       433       856      1510      1847       281
```

Flower colour

There are five sources of information for flower colour, but it must be cleaned before colors can be merged.

```
   Baseflor        BIEN        GIFT FlorealData  YvozValPol
        253        2096        2093        1746        1202
```

```
        FlorealData
bien     Blanc Bleu Jaune Marron Noir Pourpre Purpurine Rose Rouge Vert Violet
  Blue       0    3     0      0    0       0         0    0     0    0      3
  Brown      0    0     0      1    0       0         0    1     0    1      0
  Green      4    0     0      2    0       0         0    2     0    3      0
  Orange     0    0     0      0    0       0         0    0     0    0      0
  Pink       0    0     0      0    0       0         0    3     0    0      0
  Purple     2   10     1      1    0       2         0    5     0    3      0
  Red        0    0     0      0    0       0         0    3     1    0      0
  White     18    0     1      0    0       0         0    2     0    0      0
  Yellow     0    0    25      0    0       0         0    0     0   39      0
```

```
baseflor
            Blanc       Blanc, jaune Blanc, jaune, bleu Blanc, jaune, rose
              428                 37                  5                  4
      Blanc, rose  Blanc, vert, rose               Bleu        Bleu, blanc
               56                  1                172                 11
 Bleu, blanc, rose        Bleu, jaune  Bleu, jaune, rose         Bleu, rose
               12                  5                  2                 14
            Jaune        Jaune, rose        Jaune, vert             Marron
              548                  3                  1                 38
             Noir               Rose               Vert         Vert, bleu
                2                354                306                 18
 Vert, jaune, rose         Vert, rose
                1                 39
```

Dispersal mode

There are three sources of information for dispersal mode, but it must be harmonized before they can be merged.

```
Lososova2023       GIFT_1       GIFT_2     Baseflor
         147         1161         1891          183


Lososova2023
                  Anemochory                  Dyszoochory
                         343                           24
                Endozoochory                  Epizoochory
                         121                           96
                  Hydrochory Local non-specific dispersal
                           8                         1428
                Myrmecochory
                         215


GIFT
   Anemochorous Anthropochorous     Autochorous  Endozoochorous   Epizoochorous
            176              75              82              29              20
   Hydrochorous  Myrmecochorous   Unspecialized      Zoochorous
             26               3              15              75


Baseflor
               Anémochore                 Autochore                 Barochore
                      556                        36                       645
             Dyszoochore              Endozoochore Endozoochore, dyszoochore
                       10                       138                         9
             Épizoochore                Hydrochore              Myrmécochore
                      643                        56                        71
```

Inflorescence

There are two sources of information for inflorescence. GIFT database considers 11 categories, while Baseflor considers 39 categories. So the categories must be harmonized before merging.

```
    GIFT Baseflor
    2092      186


GIFT
      Catkin         Cyme      Panicle Pseudoraceme       Raceme        Spike
           6           23          152            3           41           53
       Umbel
          42


Baseflor
               Capitule de capitules                      Capitule simple
                                  11                                   35
                                Cône                             Corymbe
                                   8                                   43
                Corymbe de capitules                             Cyathe
                                  54                                   36
                          Cyme bipare                    Cyme biscorpioïde
                                 116                                    9
                   Cyme capituliforme                         Cyme d'épis
                                  27                                    7
                       Cyme d'ombelles                    Cyme de capitules
                                  27                                   28
                   Cyme de glomérules                       Cyme multipare
                                  11                                   42
               Cyme unipare hélicoïde              Cyme unipare scorpioïde
                                  99                                    6
                       Épi d'épillets                     Épi de capitules
                                 106                                   27
               Épi de cymes triflores                          Épi simple
                                  65                                   89
            Fleur solitaire latérale            Fleur solitaire terminale
                                  39                                   17
                          Glomérules                Glomérules spiciformes
                                 135                                   28
                 Ombelle d'ombellules                       Ombelle simple
                                  94                                   54
                Ombelle simple d'épis                   Panicule d'épillets
```

6
Panicule spiciforme
16
Racème d'épis
30
Racème de capitules
175
Racème de cymes unipares hélicoïdes
31
Racème de racèmes
40
Spadice
7
175
Racème capituliforme
96
Racème d'ombelles
4
Racème de cymes bipares
16
Racème de cymes unipares scorpioïdes
39
Racème simple
355

Lifecycle

There are two sources of information for lifecycle. The categories are quite coherent among GIFT and Hodgson2023 if `Herb.peren` and `Woody.peren` are merged into `Perennial`.

```
Hodgson2023        GIFT
       1347         314
```

```
             GIFT
Hodgson2023   Annual Biennial Perennial Variable <NA>
  Ann            470        2         7        0   61
  Bienn            1       29         1        0   25
  Herb.perenn      1        2       367        0   24
  Woody.peren      0        0        13        0    0
  <NA>           386       33       699        2  293
```

Clonal index

There are two sources of information for clonal index (Ladouceur 2019 and CLOPLA) but most of them are incomplete. The information seems coherent among the two databases. The order of preference is Ladouceur 2019 then CLOPLA.

```
Ladouceur2019        CLOPLA        filled
         2176          1845          1835
```

```
             CLOPLA
Ladouceur2019    1    2    3    4    5    6    7 <NA>
         2       0   13    0    0    0    0    0    3
         3       0    0   56    1    0    0    0    9
         4       0    0    1   70    0    0    0    8
         5       0    0    0    0   42    0    0    8
         6       0    0    0    0    0    9    0    4
         <NA>    3   11   86   99   99   26    7 1861
```